



EXPERIMENT-8

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Semester: 6th

Subject Name: JavaLab

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Date of Performance:

Subject Code: 22CSH-359

AIM: Develop web applications using Servlets and JSP for user input handling, database integration.

Objective: To develop a **web application** using **Servlets** and **JavaServer Pages (JSP)** that facilitates **user input handling** and integrates with a **database** for dynamic content management and storage. The project will cover key aspects such as user interface design, handling form submissions, processing user data, performing database operations (CRUD), and displaying results on a webpage.

Problem 1: Write a servlet to accept user credentials through an HTML form and display a personalized welcome message if the login is successful.

Code:

HTML Form (login.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Login</title>
</head>
<body>
  <h2>Login Form</h2>
  <form action="login" method="POST">
    <label for="username">Username:</label><br>
    <input type="text" id="username" name="username" required><br><br>
    <label for="password">Password:</label><br>
    <input type="password" id="password" name="password" required><br><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

Java Servlet (LoginServlet.java)

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class LoginServlet extends HttpServlet {

    // Hardcoded credentials for demonstration purposes
    private static final String VALID_USERNAME = "user123";
```

```
private static final String VALID_PASSWORD = "password123";
```

```
@Override
```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response) throws  
ServletException, IOException {
```

```
    // Retrieve username and password from the form
```

```
    String username = request.getParameter("username");
```

```
    String password = request.getParameter("password");
```

```
    // Set content type for the response
```

```
    response.setContentType("text/html");
```

```
    // Get the output writer
```

```
    PrintWriter out = response.getWriter();
```

```
    // Validate user credentials
```

```
    if (VALID_USERNAME.equals(username) && VALID_PASSWORD.equals(password)) {
```

```
        // If credentials are valid, display a personalized welcome message
```

```
        out.println("<html><body>");
```

```
        out.println("<h1>Welcome, " + username + "!</h1>");
```

```
        out.println("<p>You have successfully logged in.</p>");
```

```
        out.println("</body></html>");
```

```
    } else {
```

```
        // If credentials are invalid, show an error message
```

```
        out.println("<html><body>");
```

```
        out.println("<h1>Login Failed</h1>");
```

```
        out.println("<p>Invalid username or password. Please try again.</p>");
```

```
        out.println("</body></html>");
```

```
    }
```

```
}
```

```
}
```

Web Deployment Descriptor (web.xml)

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<web-app xmlns="http://java.sun.com/xml/ns/javaee"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
```

```
        http://java.sun.com/xml/ns/javaee/web-app_3_0.xsd"
```

```
    version="3.0">
```

```
<servlet>
```

```
    <servlet-name>LoginServlet</servlet-name>
```

```
    <servlet-class>LoginServlet</servlet-class>
```

```
</servlet>
```

```
<servlet-mapping>
```

```
    <servlet-name>LoginServlet</servlet-name>
```

```
    <url-pattern>/login</url-pattern>
```

```
</servlet-mapping>
```

```
</web-app>
```

Output:

```
Welcome, user123!
```

```
You have successfully logged in.
```

Login Failed

Invalid username or password. Please try again.

Problem 2: Create a servlet integrated with JDBC to display a list of employees from a database. Include a search form to fetch employee details by ID.

Code:

Database Setup

```
CREATE DATABASE company_db;
```

```
USE company_db;
```

```
CREATE TABLE employees (  
    id INT PRIMARY KEY,  
    name VARCHAR(100),  
    position VARCHAR(100),  
    department VARCHAR(100),  
    salary DECIMAL(10, 2)  
);
```

```
INSERT INTO employees (id, name, position, department, salary)  
VALUES
```

```
(1, 'Alice Johnson', 'Software Engineer', 'IT', 80000.00),  
(2, 'Bob Smith', 'Manager', 'HR', 95000.00),  
(3, 'Charlie Lee', 'Data Analyst', 'Analytics', 75000.00),  
(4, 'David Kim', 'Developer', 'IT', 70000.00);
```

HTML Form (employee_search.html)

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
    <title>Employee Search</title>  
</head>  
<body>  
    <h2>Employee Search</h2>  
    <form action="EmployeeServlet" method="POST">  
        <label for="employeeId">Employee ID:</label>  
        <input type="text" id="employeeId" name="employeeId" required>  
        <input type="submit" value="Search">  
    </form>  
    <h3>Employee List</h3>  
    <table border="1">  
        <tr>  
            <th>ID</th>  
            <th>Name</th>  
            <th>Position</th>  
            <th>Department</th>  
            <th>Salary</th>  
        </tr>  
    </table>  
</body>  
</html>
```

JDBC Servlet (EmployeeServlet.java)

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class EmployeeServlet extends HttpServlet {

    private static final String JDBC_URL = "jdbc:mysql://localhost:3306/company_db";
    private static final String JDBC_USER = "root"; // change to your DB username
    private static final String JDBC_PASSWORD = "password"; // change to your DB password

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        String employeeId = request.getParameter("employeeId");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        try {
            // Connect to the database
            Connection connection = DriverManager.getConnection(JDBC_URL, JDBC_USER,
JDBC_PASSWORD);

            // Prepare the SQL query to search for a specific employee by ID
            String query = "SELECT * FROM employees WHERE id = ?";
            PreparedStatement stmt = connection.prepareStatement(query);
            stmt.setInt(1, Integer.parseInt(employeeId));
            ResultSet rs = stmt.executeQuery();

            // Display search results
            out.println("<html><body>");
            out.println("<h2>Search Result</h2>");
            if (rs.next()) {
                out.println("<table border='1'>");

                out.println("<tr><th>ID</th><th>Name</th><th>Position</th><th>Department</th><th>Salary</th><
/tr>");
                out.println("<tr>");
                out.println("<td>" + rs.getInt("id") + "</td>");
                out.println("<td>" + rs.getString("name") + "</td>");
                out.println("<td>" + rs.getString("position") + "</td>");
                out.println("<td>" + rs.getString("department") + "</td>");
                out.println("<td>" + rs.getDouble("salary") + "</td>");
                out.println("</tr>");
                out.println("</table>");
            } else {
                out.println("<p>No employee found with ID " + employeeId + "</p>");
            }

            // Close connections
            rs.close();
            stmt.close();
            connection.close();
            out.println("</body></html>");
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

```

    } catch (SQLException e) {
        out.println("<p>Error: " + e.getMessage() + "</p>");
    }
}

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    try {
        // Connect to the database
        Connection connection = DriverManager.getConnection(JDBC_URL, JDBC_USER,
JDBC_PASSWORD);

        // SQL query to fetch all employees
        String query = "SELECT * FROM employees";
        Statement stmt = connection.createStatement();
        ResultSet rs = stmt.executeQuery(query);

        out.println("<html><body>");
        out.println("<h2>All Employees</h2>");
        out.println("<table border='1'>");

        out.println("<tr><th>ID</th><th>Name</th><th>Position</th><th>Department</th><th>Salary</th><
/tr>");
        while (rs.next()) {
            out.println("<tr>");
            out.println("<td>" + rs.getInt("id") + "</td>");
            out.println("<td>" + rs.getString("name") + "</td>");
            out.println("<td>" + rs.getString("position") + "</td>");
            out.println("<td>" + rs.getString("department") + "</td>");
            out.println("<td>" + rs.getDouble("salary") + "</td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("</body></html>");

        // Close connections
        rs.close();
        stmt.close();
        connection.close();
    } catch (SQLException e) {
        out.println("<p>Error: " + e.getMessage() + "</p>");
    }
}
}

```

Web Deployment Descriptor (web.xml)

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://java.sun.com/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
        http://java.sun.com/xml/ns/javaee/web-app_3_0.xsd"
    version="3.0">
    <servlet>

```

```

        <servlet-name>EmployeeServlet</servlet-name>
        <servlet-class>EmployeeServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>EmployeeServlet</servlet-name>
        <url-pattern>/EmployeeServlet</url-pattern>
    </servlet-mapping>
</web-app>

```

Output:

Displaying Employee List:

Upon visiting the servlet URL (e.g., `http://localhost:8080/yourAppName/EmployeeServlet`), the page will display all employees in a table:

ID	Name	Position	Department	Salary
1	Alice Johnson	Software Engineer	IT	80000.00
2	Bob Smith	Manager	HR	95000.00
3	Charlie Lee	Data Analyst	Analytics	75000.00
4	David Kim	Developer	IT	70000.00

Searching for an Employee by ID:

Search Result:

ID	Name	Position	Department	Salary
1	Alice Johnson	Software Engineer	IT	80000.00

Problem 3: Develop a JSP-based student portal. Include a form for entering attendance details and save them to the database using a servlet.

Code:

Database Setup

```
CREATE DATABASE student_portal;
```

```
USE student_portal;
```

```

CREATE TABLE attendance (
    id INT AUTO_INCREMENT PRIMARY KEY,
    student_id INT NOT NULL,
    student_name VARCHAR(100),
    date DATE,
    status VARCHAR(20),
    FOREIGN KEY (student_id) REFERENCES students(id)
);

```

```
CREATE TABLE students (
```

```

    id INT PRIMARY KEY,
    name VARCHAR(100)
);

-- Insert some students for testing
INSERT INTO students (id, name) VALUES (1, 'Alice'), (2, 'Bob'), (3, 'Charlie');

```

JSP Form (attendance_form.jsp)

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Attendance Form</title>
</head>
<body>
    <h2>Enter Attendance</h2>
    <form action="SubmitAttendanceServlet" method="POST">
        <label for="student_id">Student ID:</label><br>
        <select name="student_id" id="student_id">
            <option value="1">1 - Alice</option>
            <option value="2">2 - Bob</option>
            <option value="3">3 - Charlie</option>
        </select><br><br>

        <label for="date">Date:</label><br>
        <input type="date" id="date" name="date" required><br><br>

        <label for="status">Status:</label><br>
        <select name="status" id="status">
            <option value="Present">Present</option>
            <option value="Absent">Absent</option>
        </select><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>

```

Servlet to Save Attendance (SubmitAttendanceServlet.java)

```

import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class SubmitAttendanceServlet extends HttpServlet {

    // Database credentials
    private static final String JDBC_URL = "jdbc:mysql://localhost:3306/student_portal";
    private static final String JDBC_USER = "root"; // Your DB username
    private static final String JDBC_PASSWORD = "password"; // Your DB password

    @Override

```

```

protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    // Get form parameters
    int studentId = Integer.parseInt(request.getParameter("student_id"));
    String date = request.getParameter("date");
    String status = request.getParameter("status");

    // Set content type for the response
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    try {
        // Connect to the database
        Connection connection = DriverManager.getConnection(JDBC_URL, JDBC_USER,
JDBC_PASSWORD);

        // Prepare SQL query to insert attendance record
        String query = "INSERT INTO attendance (student_id, date, status) VALUES (?, ?, ?)";
        PreparedStatement stmt = connection.prepareStatement(query);
        stmt.setInt(1, studentId);
        stmt.setDate(2, Date.valueOf(date)); // Convert String to Date
        stmt.setString(3, status);

        // Execute the query
        int rowsAffected = stmt.executeUpdate();

        if (rowsAffected > 0) {
            // Display success message
            out.println("<html><body>");
            out.println("<h2>Attendance recorded successfully for student ID " + studentId + " on " + date
+ ".</h2>");
            out.println("<a href='attendance_form.jsp'>Go back to form</a>");
            out.println("</body></html>");
        } else {
            // Handle failure case
            out.println("<html><body>");
            out.println("<h2>Error while recording attendance. Please try again.</h2>");
            out.println("<a href='attendance_form.jsp'>Go back to form</a>");
            out.println("</body></html>");
        }

        // Close connections
        stmt.close();
        connection.close();
    } catch (SQLException e) {
        out.println("<html><body>");
        out.println("<h2>Error: " + e.getMessage() + "</h2>");
        out.println("<a href='attendance_form.jsp'>Go back to form</a>");
        out.println("</body></html>");
    }
}
}

```

Web Deployment Descriptor (web.xml)

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://java.sun.com/xml/ns/javaee"

```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
http://java.sun.com/xml/ns/javaee/web-app_3_0.xsd"
version="3.0">
<servlet>
  <servlet-name>SubmitAttendanceServlet</servlet-name>
  <servlet-class>SubmitAttendanceServlet</servlet-class>
</servlet>
<servlet-mapping>
  <servlet-name>SubmitAttendanceServlet</servlet-name>
  <url-pattern>/SubmitAttendanceServlet</url-pattern>
</servlet-mapping>
</web-app>
```

Output:

Attendance Form:

```
Enter Attendance
Student ID: [1 - Alice, 2 - Bob, 3 - Charlie]
Date: [Select Date]
Status: [Present, Absent]
[Submit Button]
```

Successful Submission:

csharp

```
Attendance recorded successfully for student ID 1 on 2023-04-01.
[Go back to form]
```

Failed Submission (e.g., Database Error):

vbnet

```
Error while recording attendance. Please try again.
[Go back to form]
```

Learning Outcome:

1. Understanding Web Application Architecture:
2. Handling User Input in Web Applications:
3. Handling User Input in Web Applications:
4. Building CRUD Operations